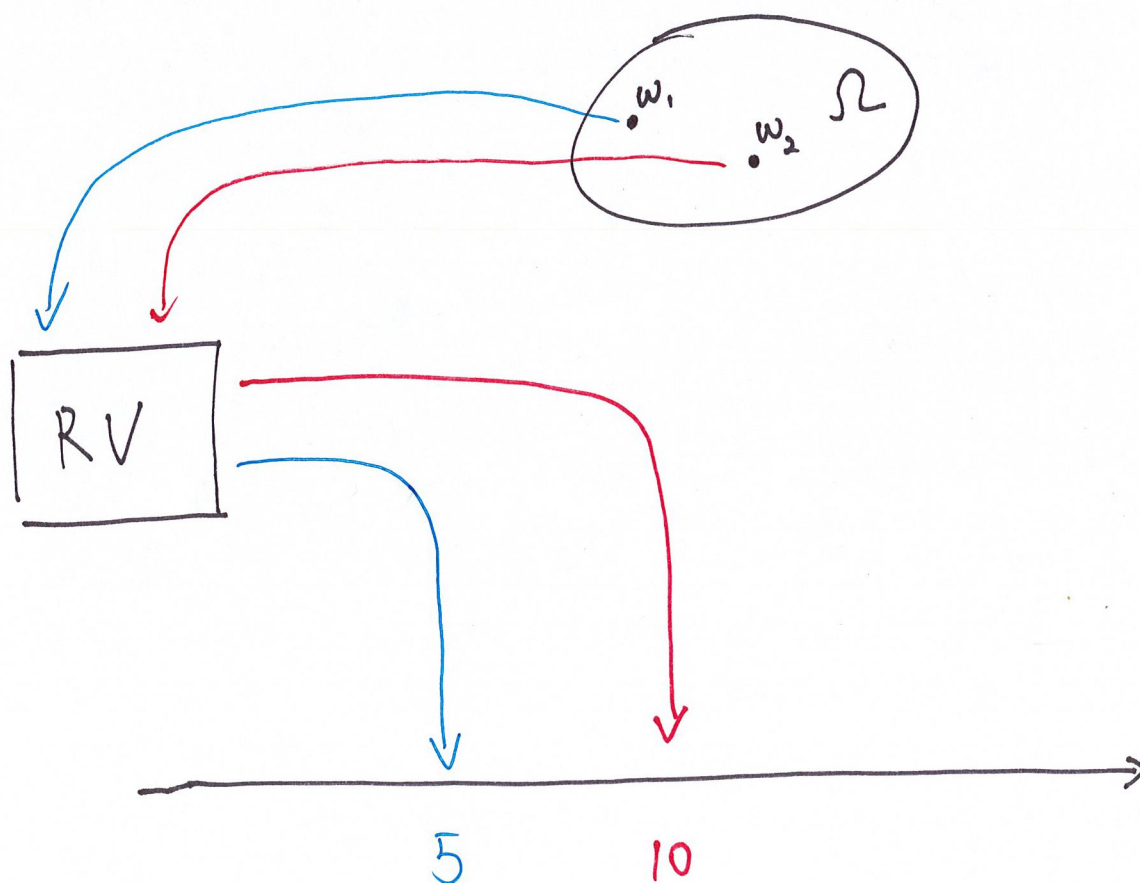


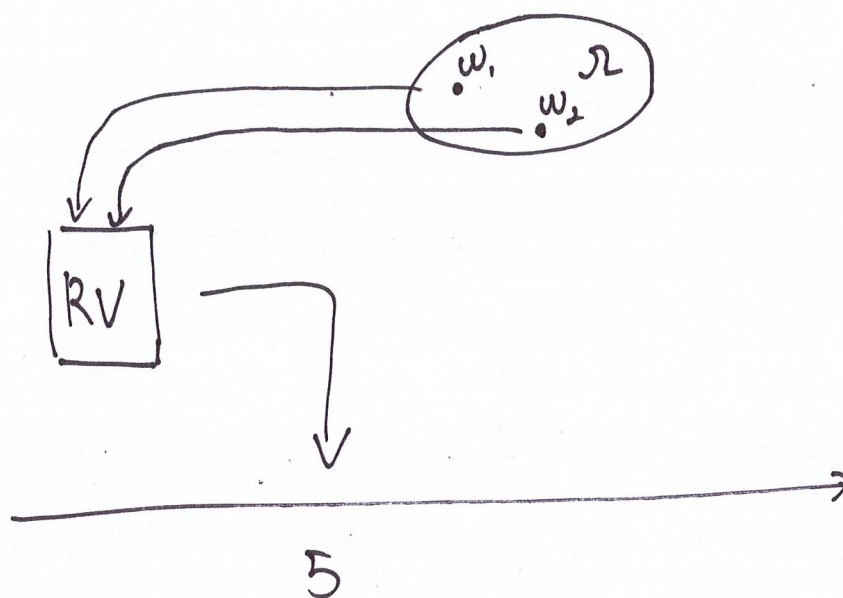
Random variables

Def. A random variable (RV) is a function from Ω to $\mathbb{R} := (-\infty, \infty)$.

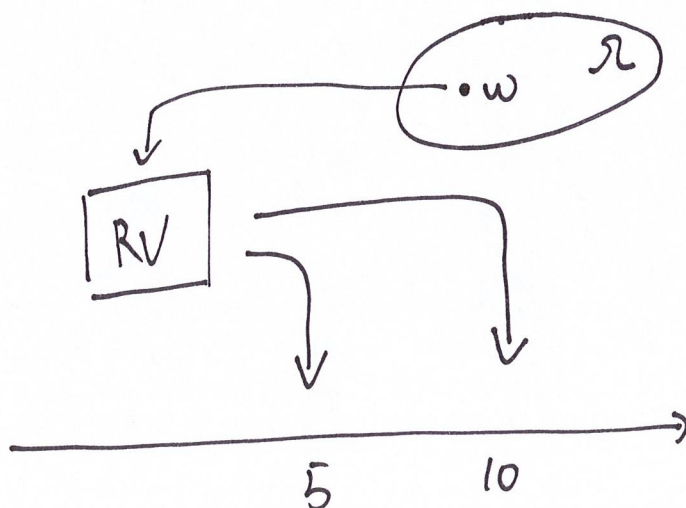
NOTE. A RV is NOT a variable nor random.



It may be possible :



It can NOT be



Def. The support of a RV is the set of all possible values.

$\text{supp}(X)$

Ex. Roll 2 6-sided dice.

- $\Omega = \{ (i, j) \mid i, j = 1, 2, \dots, 6 \}$

- Sum of the two dice :

$$X(\omega) = i + j$$

$$\text{supp}(X) = \{ 2, 3, \dots, 12 \}$$

- Maximum of the two dice :

$$Y(\omega) = \max(i, j)$$

$$\text{supp}(Y) = \{ 1, 2, \dots, 6 \}$$

▮ NOTE. For a given experiment, we can define ∞ many RV's on Ω .

Ex. Consider the genders of four new born babies.

- $\Omega = \{(s_1, s_2, s_3, s_4) \mid s_i \in \{\text{boy}, \text{girl}\}\}$
- We are interested in the gender of the first baby.
- $X(\{s_1, s_2, s_3, s_4\}) = s_1$ is NOT a RV since it does NOT output a number.
- Alternatively,

$$X(\omega) = \begin{cases} 0, & \text{if } s_1 = \text{boy} \\ 1, & \text{if } s_1 = \text{girl} \end{cases}$$

▮ Discrete RV's : The support of a discrete RV is finite $(1, 2, \dots, 6)$ or countably infinite $(1, 2, 3, \dots)$

▮ Applications :

- # of students coming to class.
- # of failures of a system.
- # of customers arriving at a store.

▮ Continuous RV's : A continuous RV takes values on continuous intervals $(0, 1)$.

▮ Applications :

- the inter-arrival time of floods.
- the price of a stock.
- the temperature of a location.